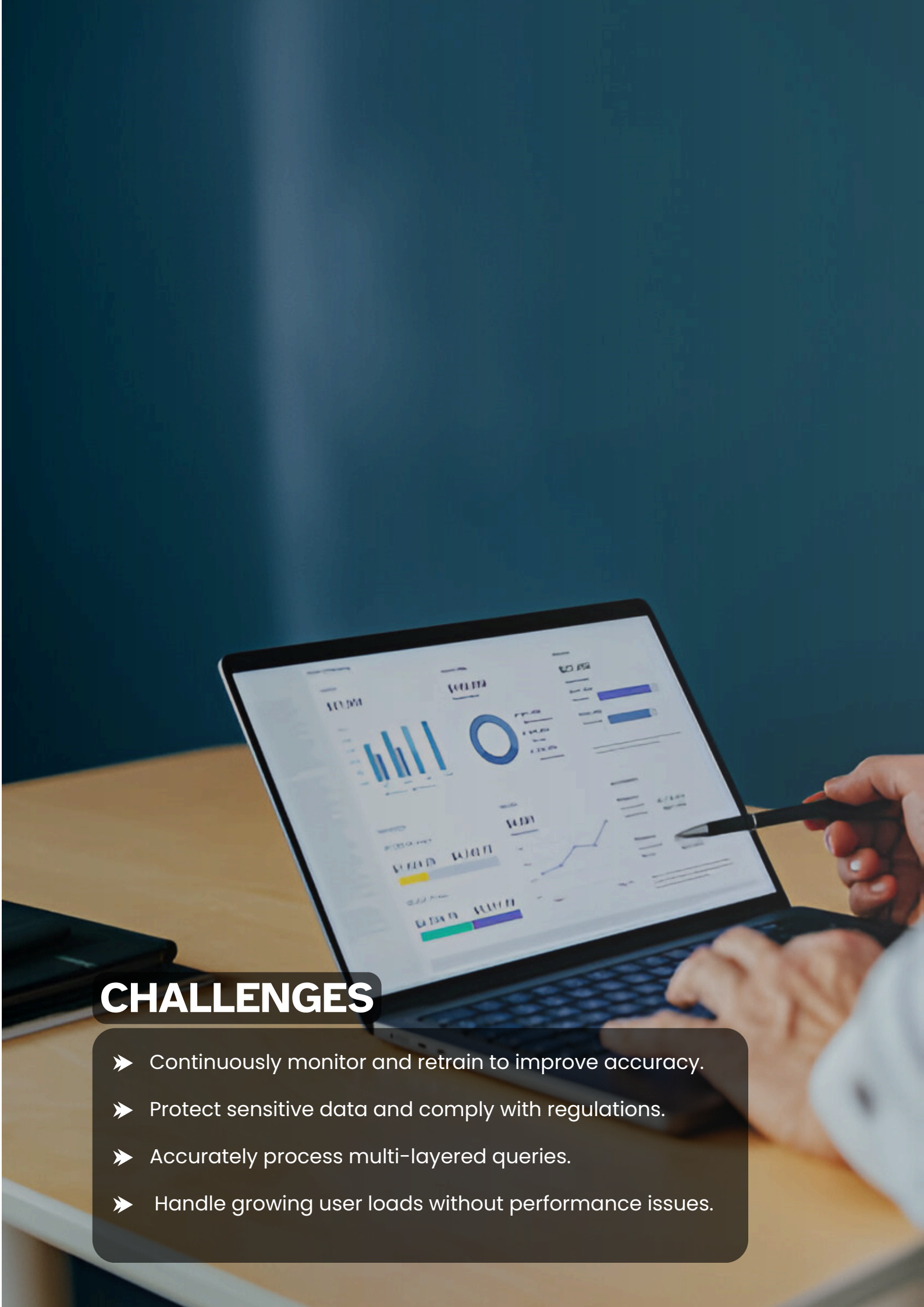


ACCELERATING BUSINESS INTELLIGENCE WITH AI-DRIVEN NATURAL LANGUAGE INSIGHTS AND INTERACTION



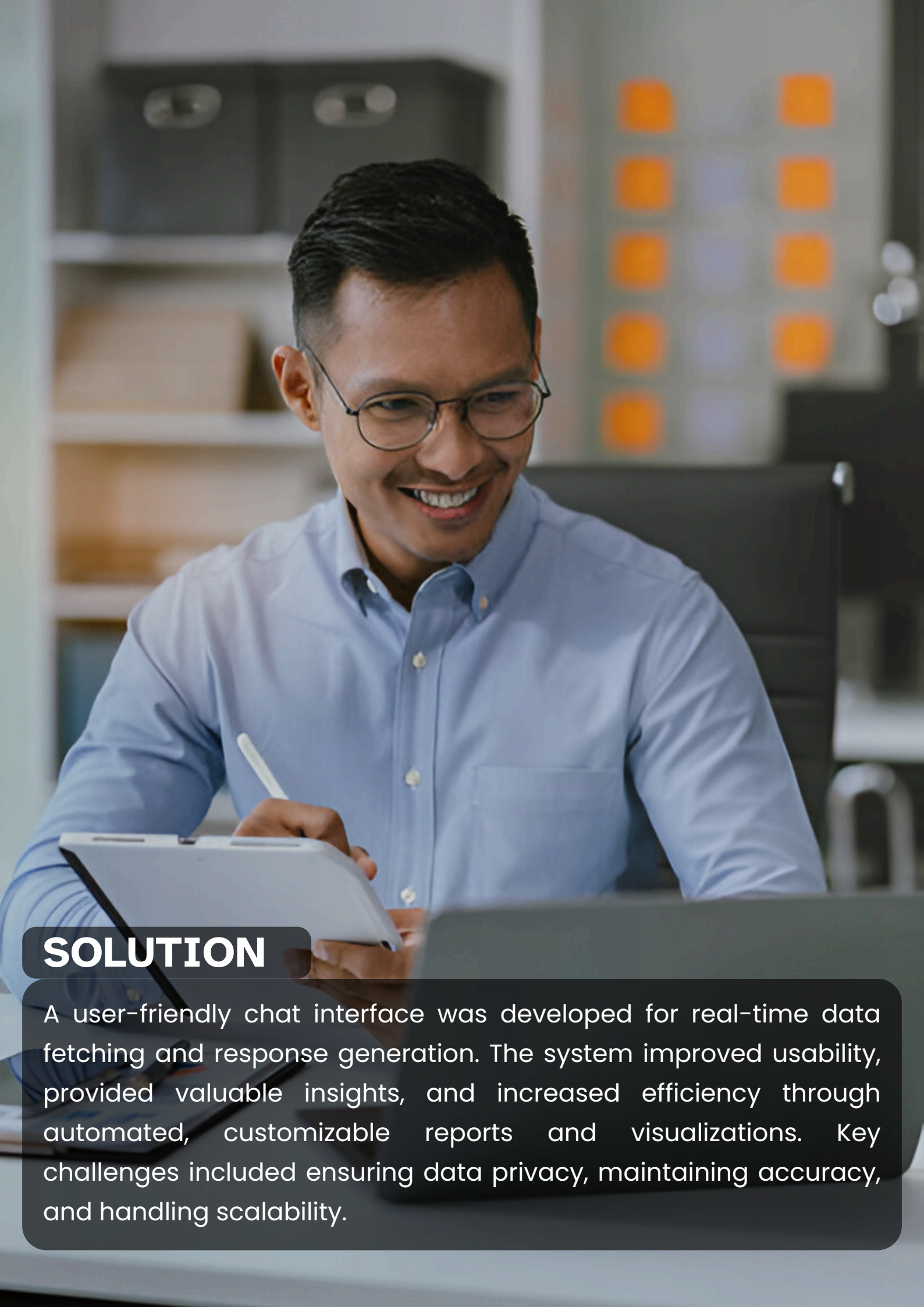
OVERVIEW

Fine-tuning a large language model (LLM) enabled seamless database interaction and automated report generation using natural language. The project involved understanding the database schema, preparing training data, and fine-tuning the LLM to generate accurate reports and respond to user queries.



CHALLENGES

- Continuously monitor and retrain to improve accuracy.
- Protect sensitive data and comply with regulations.
- Accurately process multi-layered queries.
- Handle growing user loads without performance issues.



SOLUTION

A user-friendly chat interface was developed for real-time data fetching and response generation. The system improved usability, provided valuable insights, and increased efficiency through automated, customizable reports and visualizations. Key challenges included ensuring data privacy, maintaining accuracy, and handling scalability.

KEY FEATURES



Tailored Reports

Generates tailored reports based on user-defined time ranges, filters, and grouping criteria.

Data Visualizations

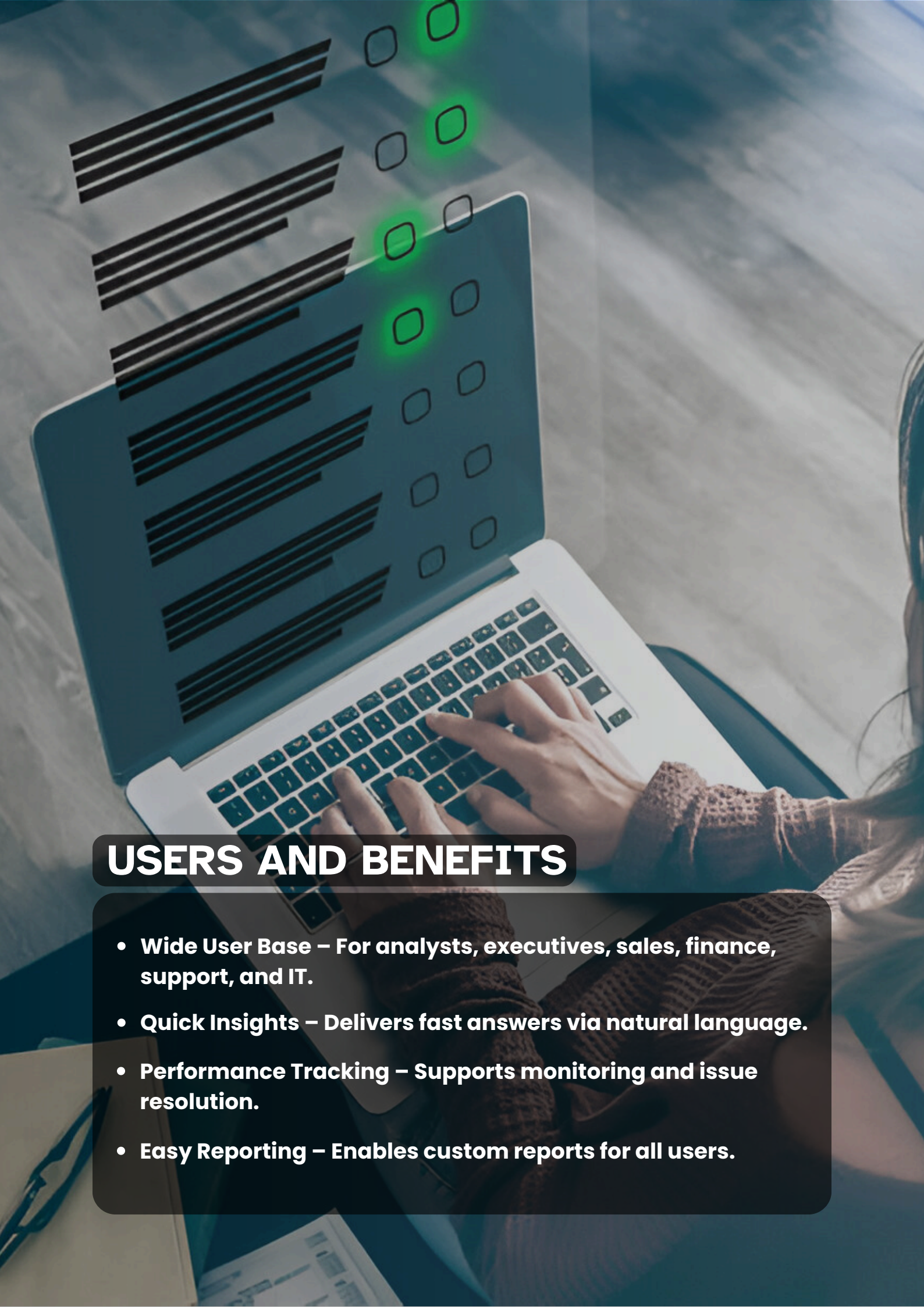
Creates charts and graphs from user queries for clear data visualization.

On-Demand Query Execution

Converts natural language queries into SQL to quickly fetch relevant data.

Real-Time Insights

Provides real-time answers and reports, enhancing decision-making speed and accuracy.



USERS AND BENEFITS

- **Wide User Base – For analysts, executives, sales, finance, support, and IT.**
- **Quick Insights – Delivers fast answers via natural language.**
- **Performance Tracking – Supports monitoring and issue resolution.**
- **Easy Reporting – Enables custom reports for all users.**



CONCLUSION

The fine-tuned LLM-based system significantly transformed how users interacted with data and generated reports. The system's ability to process natural language queries and generate real-time insights led to major improvements in usability, efficiency, and decision-making.